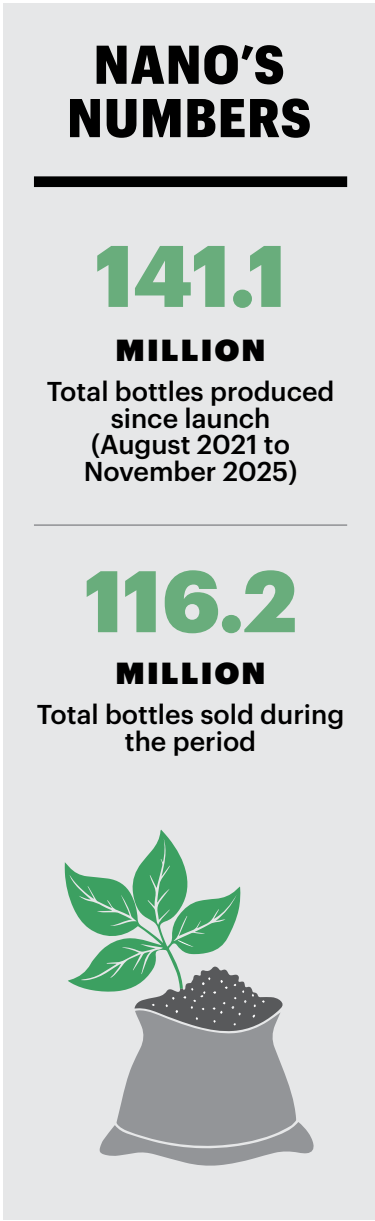


This was after Agriculture & Farmers Welfare Minister Shivraj Singh Chauhan wrote to chief ministers, in July 2025, seeking legal action against forced sale, following country-wide protests.

While the Indian Farmers Fertiliser Cooperative Ltd (IFFCO), the inventor and manufacturer of nano fertilisers, denies forced selling and assures it will implement the FCO, Coromandel International, a major private sector fertiliser company manufacturing nano DAP, says its experience has been different. Arun Alagappan, its Executive Chairman, says farmers “have used our nano fertilisers and are now swearing by it...Not only is its application easy, it’s done through drone application. So, that is something which is catching on very, very quickly in this industry.”

30 | Alagappan narrated his experience with nano DAP: “There’s been a lot of talk on nano fertilisers and we have had a very good run with it. In fact, I was recently at an institute in Hyderabad (Professor Jayashankar Telangana Agricultural University), where we’ve done a lot of trial plots with the industry. And we feel that nano is a good product. It has to be applied correctly, and there’s absolutely no problem with nano as a product.”

Nonetheless, farmers from Punjab and Haryana told *Business Today* that forced sales are continuing. “Despite the minister’s official stand, when we buy conventional fertilisers and pesticides, local agriculture cooperative societies and private vendors make it conditional on buying nano urea and nano DAP bottles,” says Parminder Singh Ghuman, a farmer from Punjab’s Sangrur.



SOURCE IFFCO

Haryana farmers’ leader Gurnam Singh Charuni confirms this: “We have been telling them, don’t impose but show results; then, farmers will buy on their own. Sometimes, farmers are forced to buy nano fertiliser because

of shortage of conventional fertilisers”. Meanwhile, the Agriculture & Farmers Welfare Ministry is evaluating the *raison d’être* of nano fertilisers.

FRESH TRIALS

On February 13, 2026, the ministry informed Parliament that all nano-fertiliser manufacturers would be required to conduct bio-efficacy trials across agro-climatic zones. Three separate mechanisms have been put in place for “evaluation” of the impact of nano fertilisers on at least 12 major crops by multiple institutions under the supervision of India Council of Agricultural Research.

These evaluations cover the entire gamut: “efficacy, utility and impact” and “the extent of replacement” of conventional fertiliser; “crop productivity and nitrogen use efficiency”; “long-term” impact on “crop yield, produce quality and soil nutrient balance as well as working mechanism of nutrient from nano fertiliser in plant metabolism”, “crop growth, soil health and nutrient uptake,” and “bio-safety/toxicology trials.”

KJ Patel, Managing Director of IFFCO, says he welcomed fresh trials because “conflicting results across different trial conditions are not an anomaly in the science of agricultural innovation. They are the very reason continued multi-location, multi-crop, multi-season trials are essential.”

Ramesh Raliya, a former IFFCO scientist credited with inventing nano urea and nano DAP, says fertiliser performance cannot be measured by a one-size-fits-all evaluation. It depends on factors like soil type,